Technology Stack

Date	27 June 2026
Team ID	xxxxxx
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	Streamlit, HTML, CSS	Streamlit was chosen to rapidly build an interactive web interface for subject selection, problem input, and results display with minimal front-end code.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python, TensorFlow, Keras, Hugging Face Transformers, PyTorch	Python powers the entire application; TensorFlow/Keras run the BiLSTM model, Hugging Face Transformers run the fine-tuned BERT model, and PyTorch supports the underlying transformer computations.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	CSV Files	CSV files were used for lightweight local storage of user interaction history, emotion results, and AI-generated guidance without requiring a dedicated database server.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Kaggle GPU (training), Local Deployment via Streamlit	Model training was performed on Kaggle GPU, while the application is currently deployed locally using Streamlit for development and demonstration; the architecture can be migrated to the cloud in future versions.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	Git, GitHub, Visual Studio Code	Git and GitHub were used for source code management, version tracking, and collaboration, with Visual Studio Code as the primary development environment.
6	Third-Party APIs / Other Tools (e.g., Stripe, Twilio, Google Maps)	Google Generative AI SDK (Gemini 2.5 Flash), Pandas, NumPy, Plotly, Scikit-learn	The Google Generative AI SDK integrates Gemini 2.5 Flash for personalized learning guidance, Pandas and NumPy handle data processing, Plotly powers the analytics dashboard, and Scikit-learn supports auxiliary ML utilities.